
Mathematical Background for the Kalam Cosmological Argument

Hilbert's Hotel, Actual Infinity, and Cantor's Set Theory

I. The Problem

Craig and Moreland's philosophical defense of the *Kalam Cosmological Argument* relies heavily on claims about the **impossibility of an actual infinite existing in reality** (pp. 181–183). To assess their arguments we must understand several core mathematical concepts:

- The distinction between **actual** and **potential infinity**
- The concept of **cardinality**
- Countable vs. uncountable infinities
- One-to-one correspondence
- Cantor's diagonal argument
- The notion of forming a set by **successive addition**

Without this mathematical framework, it is impossible to evaluate whether Craig's metaphysical conclusions genuinely follow from the mathematics.

II. Potential vs. Actual Infinity

This distinction is absolutely central.

1. Potential Infinite

A process that:

- Can continue indefinitely
- Is never complete
- Always finite at every stage

Example:

- Counting: 1, 2, 3, 4, ...
- Dividing a line segment: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, ...

At every stage, the number of elements is finite.

This is sometimes described as **infinity as a limit**.

2. Actual Infinite

A collection:

- With a definite number of members
- That number is greater than any natural number
- Complete and total

Formally:

A set is actually infinite if it can be placed in **one-to-one correspondence with a proper subset of itself**.

Example:

The set of natural numbers:

$$\{\mathbb{N}\} = \{1, 2, 3, 4, \dots\}$$

This is what Craig calls an **actual infinite** (p. 181).

The entire Hilbert's Hotel argument depends on whether such a collection can exist in reality.

III. Cardinality and One-to-One Correspondence

To understand Hilbert's Hotel, we must understand **cardinality**.

Cardinality

The size of a set.

For finite sets:

- Easy to count.

For infinite sets:

- Defined by **one-to-one correspondence** (also known as **bijection**).

Two sets have the same cardinality if:

- Every element in one can be paired with exactly one element in the other.
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Example: Even Numbers

Natural numbers:

1, 2, 3, 4, 5, ...

Even numbers:

2, 4, 6, 8, ...

Pair them:

$$n \mapsto 2n$$

Every natural number corresponds to exactly one even number.

Thus:

$$|\mathbb{N}| = |\{\text{Even numbers}\}|$$

Even though the evens are a **proper subset** of the naturals.

This violates finite intuitions about size.

This is the mathematical principle underlying **Hilbert's Hotel**.

IV. Hilbert's Hotel

Hilbert imagines a hotel with:

- Countably infinitely many rooms.
- All rooms occupied.

Key Operations:

1. Add one guest:
Shift guest in room $n \rightarrow$ room $n+1$.
Room 1 becomes empty.
2. Add infinitely many guests:
Shift guest in room $n \rightarrow$ room $2n$.
All odd-numbered rooms become empty.

Mathematically:

This works because the natural numbers are countably infinite.

Philosophical claim (Craig):

These results are **absurd in reality**.

Mathematical fact:

They are perfectly consistent within set theory.

The disagreement is not about mathematics.

It is about whether mathematical structures can be instantiated in reality.

V. Countable vs. Uncountable Infinity

Craig discusses **actual infinite collections**, but Cantor showed that not all infinities are equal.

Countably Infinite

A set is countable if it can be placed in one-to-one correspondence with $\mathbb{N}$.

Examples:

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- Natural numbers
 - Integers
 - Rational numbers

Cardinality:

$\aleph_0$

(Hebrew letter aleph-zero)

Hilbert's Hotel concerns this type.

Uncountable Infinite

Cantor proved:

The real numbers are **uncountable**.

There are strictly more real numbers than natural numbers.

This is shown through:

Cantor's Diagonal Proof

VI. Cantor's Diagonal Argument (Essential Understanding)

Purpose:

To show that the real numbers between 0 and 1 cannot be listed.

Assume:

We try to list all real numbers between 0 and 1:

0.123456...

0.987654...

0.111111...
etc.

Construct a new number:

- Take first digit of first number.
 - Change it.
- Take second digit of second number.
 - Change it.
- Continue diagonally.

This produces a new number differing from every number in the list at least at one decimal place.

Therefore:
It was not in the list.

Conclusion:
The reals are uncountable.

Why This Matters for the Kalam Argument

Craig insists:

- He is not rejecting Cantor's mathematics.
- He accepts that actual infinities are consistent in abstract mathematics.
- He denies they can exist in physical reality.

So we must understand:

Mathematical consistency $\neq$ metaphysical instantiability.

VII. Infinite Regress and Successive Addition

Craig's second philosophical argument depends on:

A collection formed by successive addition cannot be actually infinite.

Mathematically:

A sequence like:

1

1,2

1,2,3

1,2,3,4

...

Never becomes infinite.

It grows without bound, but at every stage is finite.

Craig applies this to time:

- Past events occur sequentially.
- If the past were infinite,
- The present could never be reached.

This depends on:

- Treating the past as a completed collection.
 - Accepting that time unfolds sequentially (A-theory).
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VIII. Series and Convergence (Zeno Comparison)

Critics compare Craig's reasoning to Zeno's paradoxes.

Understanding required:

Infinite series can sum to finite values.

Example:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

This converges to 1.

Thus:

Traversing infinitely many *decreasing* intervals is possible.

Craig replies:

In an infinite past, intervals are equal and actual, not decreasing and potential.

Thus we must understand:

- Convergent series
 - Difference between potential infinite division and completed infinite set
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